Peppol

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Peppol Testbed

eDelivery test suite user guide

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1 Introduction

This document provides information related to the eDelivery testing. Its purpose is to demonstrate and to provide a guide on how to use the Testbed User Interface. It describes the prerequisites as well as the key steps in carrying out the various test cases. The target audience for this document is Service Providers (SPs) that are interested in implementing Peppol and wish to test their implementations by using the Testbed environment.

1.1 Glossary

Abbreviations used in this document:

Abbreviation	Meaning
AP	Access Point
SMP	Service Metadata Publisher
SP	Service Provider
UI	User Interface

Table 1 glossary

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1.2 Process Overview

To execute a Peppol Process within the Peppol network, there are several capabilities that the SPs must have. OpenPeppol makes it easier to verify these capabilities through a structured predefined testing process.

To achieve this, a test suite for eDelivery testing is offered by Peppol, which requires the completion of six test cases that verify the SP's eDelivery deployment, configuration, and compliance to Peppol specifications of Access Point (AP).

2 Prerequisites and preparation

2.1 Prerequisites

SPs must meet the following requirements before doing any testing.

2.1.1 Peppol PKI Test Certificate

SPs are authenticated to the Testbed using PKI client authentication. As a result, the SP must have obtained a Peppol PKI test certificate and have it imported in their browser's keystore.

2.1.2 Access Point deployment

SPs must have an AP deployed, as a result they need the following:

- AS4 URL available – must refer to HTTPS
- AP must be accessible over the internet
- AP must implement HTTPS with certificate chains to Certificate Authorities (CAs) which are trusted by Peppol.
- AP must have installed the same Peppol PKI AP test certificate used for the authentication to the Testbed

2.2 Supported browsers

We can only guarantee that the Testbed website behaves in the correct manner if you are using any of the below web browsers:

- Google Chrome, version 131.0 and above
- Firefox, version 133.0.3 and above
- Microsoft Edge, version 132.0 and above

2.3 Preparation

A Peppol PKI AP test certificate must be installed in your browser before visiting the Testbed for the first time.

Client certificate authentication must be setup in your browser according to the following instructions:

Windows Google Chrome

Windows Mozilla Firefox

OS-X Generic

After installing the PKI certificates, go to https://www.testbed.peppol.org, which is the URL of the Testbed environment.

The browser will prompt you to identify yourself with a certificate; choose your test certificate.

The Testbed includes a set of available test suites that are enabled or disabled based on the certificate type imported by the user. You will only have access to test suites compatible with the currently imported certificate. In case of needing to use a disabled test suite (one that is not enabled by your current imported certificate type) you need to clear the cookies in your browser to reset your session and import a certificate that corresponds to the test suite you want to use.

Instructions on clearing browser cookies can be found here:

Windows Mozilla Firefox

Windows Google Chrome

Windows Microsoft Edge

3 Testbed eDelivery test suite user interface

The eDelivery test suite is used to help evaluate new AP deployments and ensure compliance with the Peppol eDelivery Network specifications. It is based on self-initiated tests that an end-user of the SP system chooses to run. The SP end user enrolls in the eDelivery test suite, selects which test cases to run, and reviews the results from the user interface.

To create an eDelivery test suite configuration, a user must first select "*ENTER eDELIVERY TEST SUITE*" in the landing page, as illustrated in Figure 1.

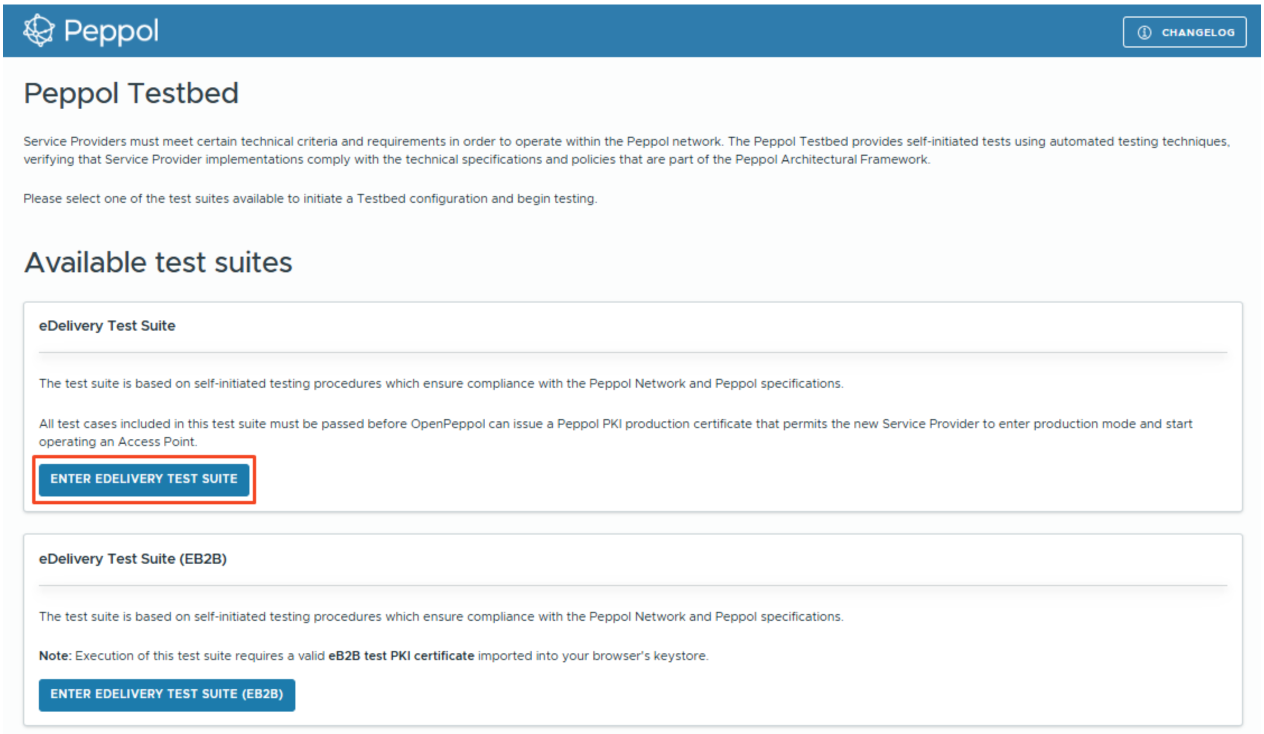

Figure 1 Testbed landing page

When the eDelivery test configuration is created, the user is navigated to the documentation page as shown in

Figure 2. The next section provides a general overview of the UI.

The UI consists of the following pages (as shown in

Figure 2):

1. eDelivery Testing > Standalone AP
2. Documentation
3. Profile

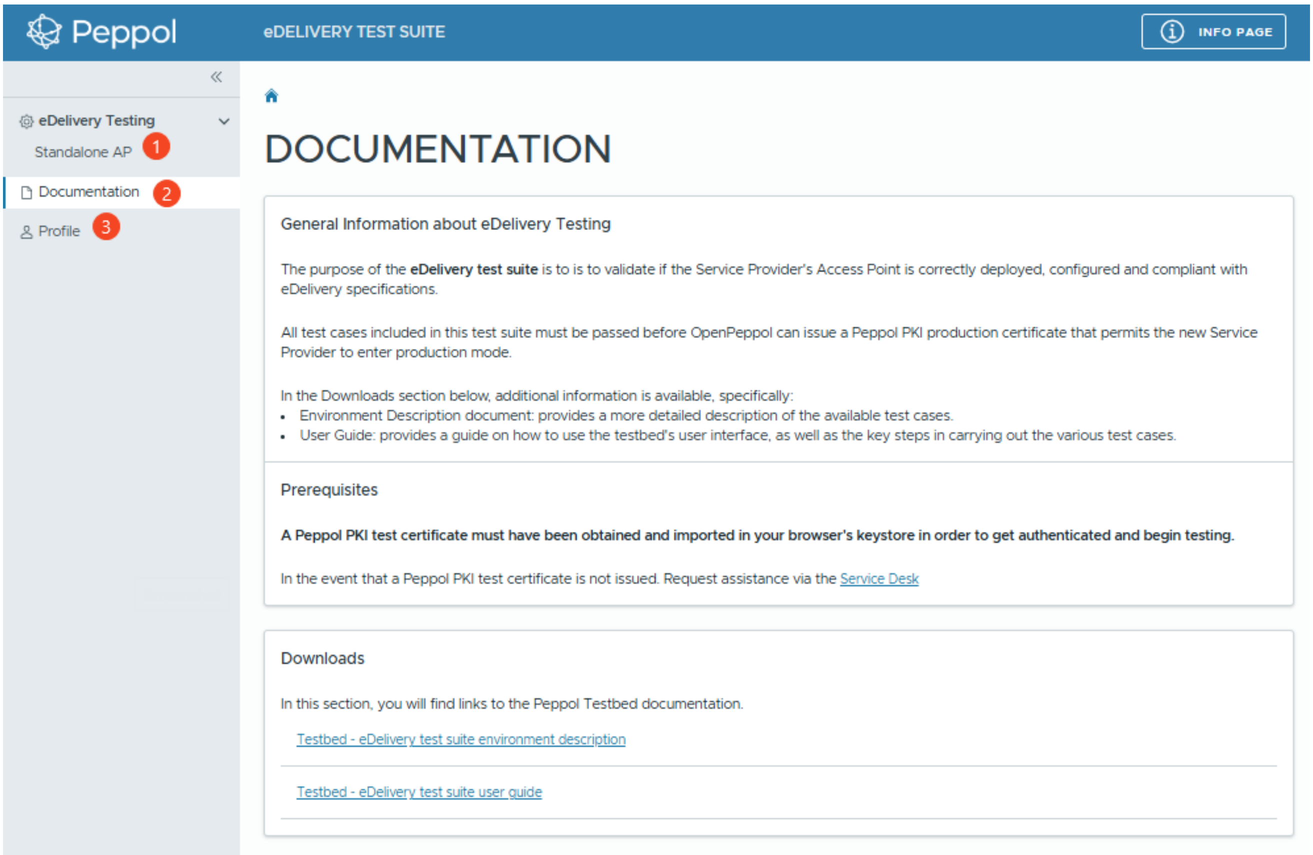

Figure 2 UI overview after entering the eDelivery test suite (documentation page)

1. The eDelivery Testing > Standalone AP page (shown in Figure 3) is used to enroll to the eDelivery test suite.

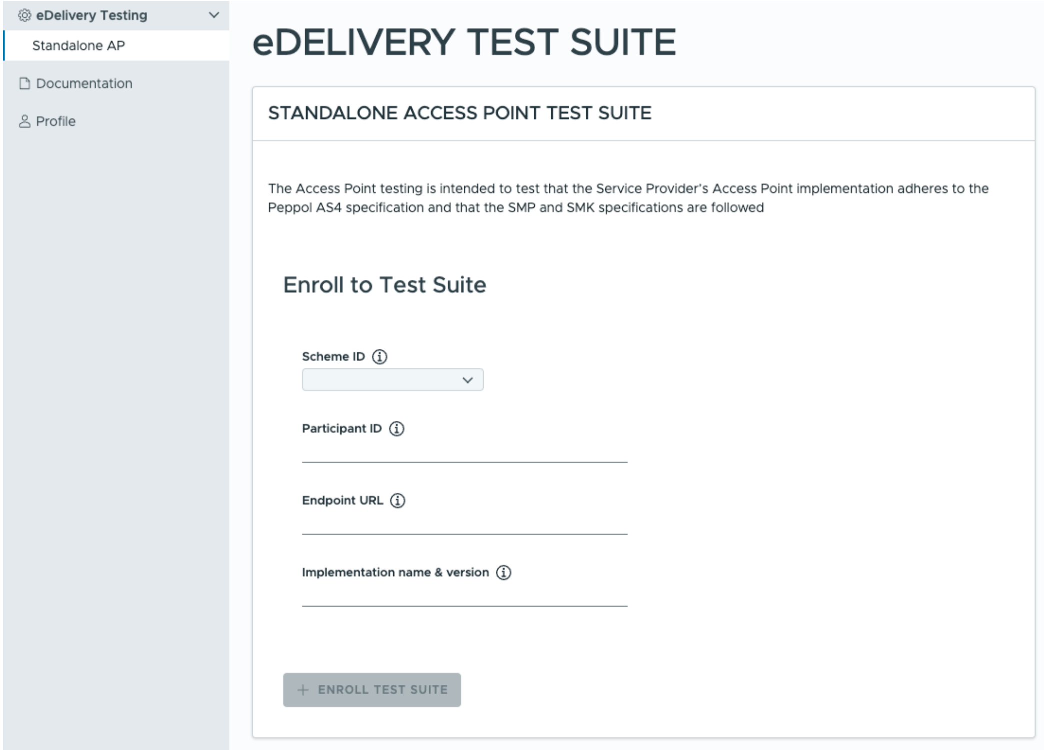

Figure 3 eDelivery test suite enrollment

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After enrolling to a test suite, the user can choose test cases to execute, as shown in Figure 4, label 1. As illustrated in Figure 4, label 2, the user may also view details about the enrolled configuration.

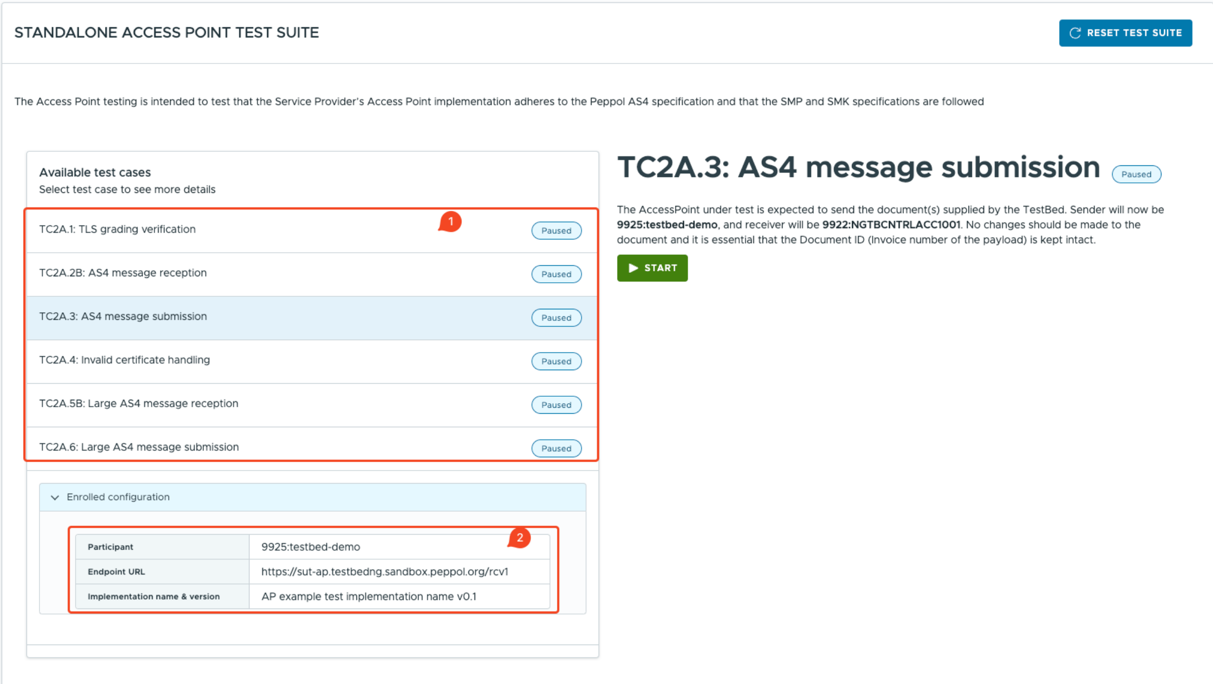

Figure 4 example of eDelivery test case selection

2. The Documentation page (shown in
3. Figure 5) collects and makes available useful links and resources (e.g., environment documentation, user guide, etc.)

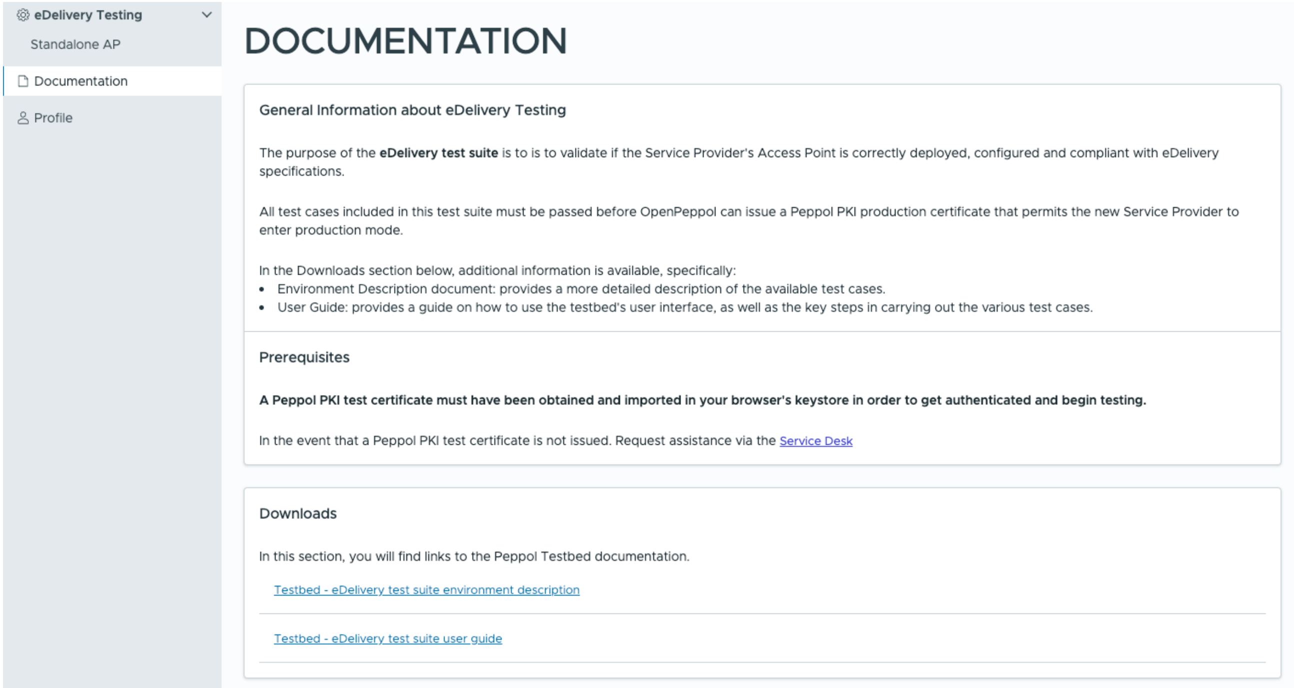

Figure 5 eDelivery test suite Documentation page

4. The Profile page (shown in Figure 6) provides details about the user account that the Testbed creates using the Peppol PKI Test Certificate that was imported into the browser.

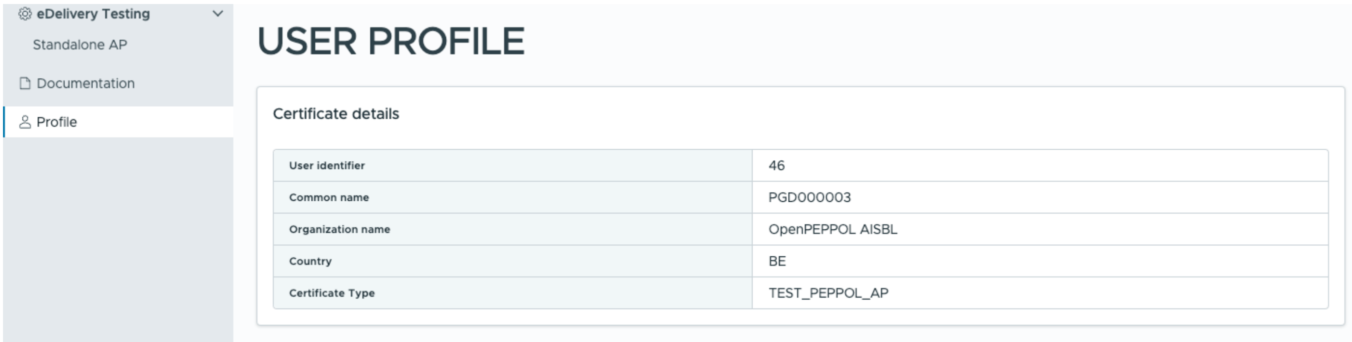

Figure 6 User profile page

4 eDelivery Test Suite Walkthrough

This part aims to show you how to use and complete the eDelivery test suite. This part requires that you have correctly setup your browser by installing a Peppol PKI test certificate as specified in section 2.3.

4.1 eDelivery test suite

Navigate to https://www.testbed.peppol.org/ after installing a test AP certificate in your browser and select the eDelivery test suite from the landing page, as shown in Figure 1. The Testbed environment will automatically recognize that it is an AP certificate and load the eDelivery test suite page if your browser is set correctly and the certificate is installed.

If this is your first time using the eDelivery test suite, you must first enroll by providing the following information:

Participant ID and Scheme ID: Give a valid participant identifier. All the test cases available in the test suite will be carried out by using these identifiers

Endpoint URL: The participant's AS4 endpoint

Implementation name & version: Give a name and version for your AP implementation

After filling in the information, click the "ENROLL TEST SUITE" button.

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An example filled-in the enrolment form is shown in Figure 7.

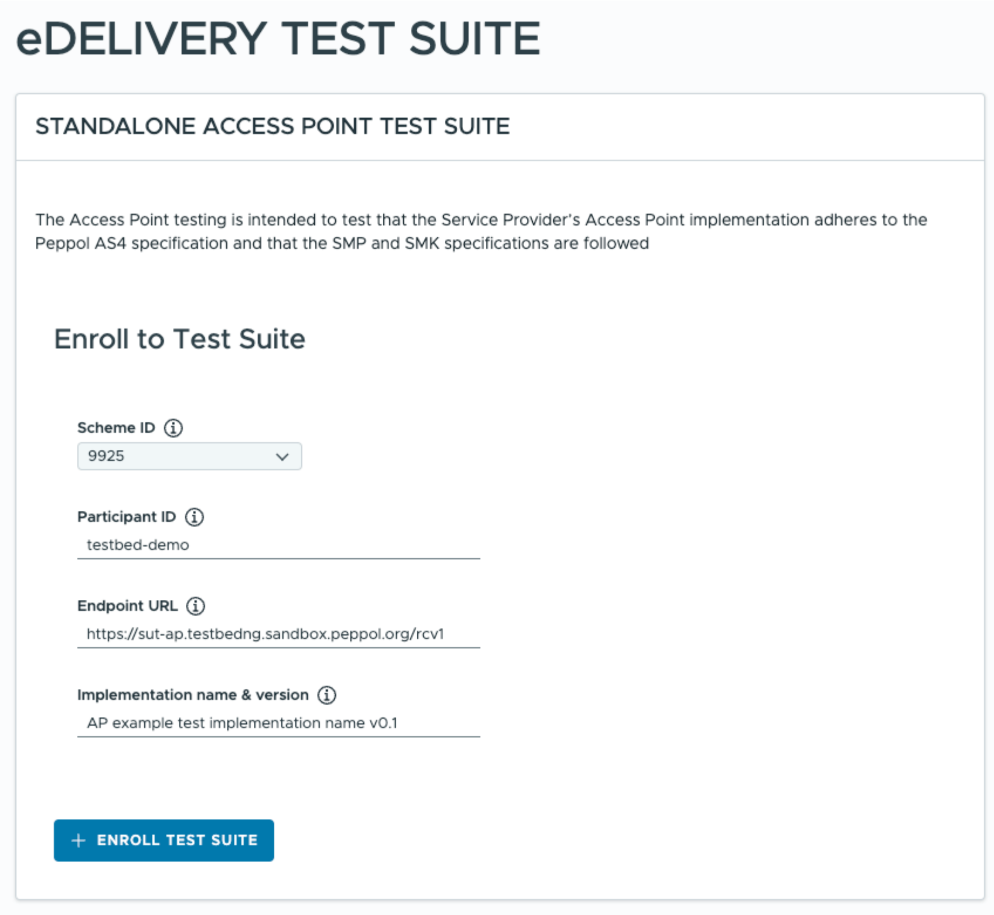

Figure 7 example AP filled in form

You will be presented with the available test cases for this test suite once you have successfully enrolled in the test suite. Figure 8 illustrates this. The following are the test cases included in the eDelivery test suite:

- TLS grading verification
- AS4 message reception
- AS4 message submission
- Invalid certificate handling
- Large AS4 message reception
- Large AS4 message submission

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It’s worth noting that the eDelivery test suite’s test cases don’t have to be performed in order. Instead, any test case is ready to be run at any moment. The sole restriction is that **only one test case may run at a time**.

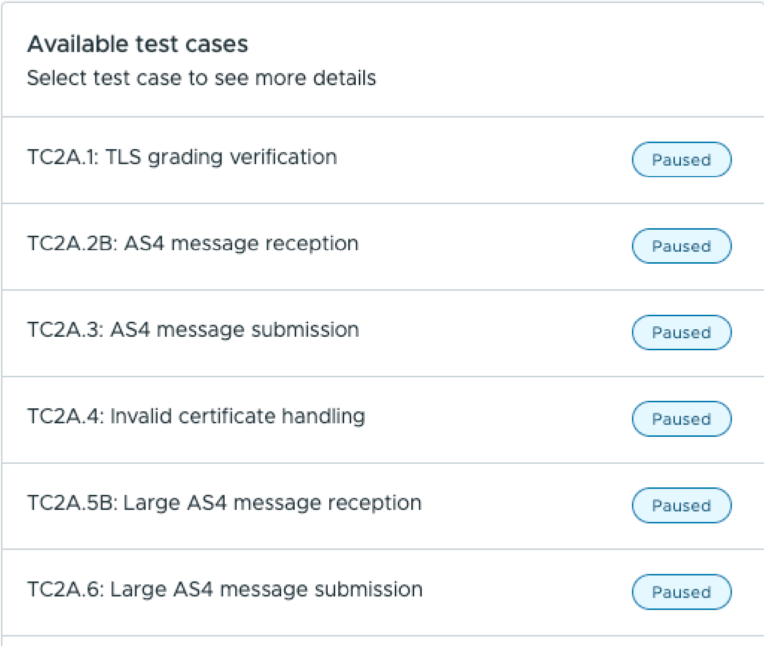

Figure 8 eDelivery test suite – available test cases

When conducting each test case, make sure to follow the test case description. You can start a test case by pressing the “Start” button showcased in Figure 9, label 3. The configuration created by the user during the test suite enrolment is shown in Figure 9, label 4. As illustrated in Figure 9, label 5, audit logs of events are available that comprise the timing of the occurrence, the action done, a description and a status. Finally, if you wish to re-enroll in a new eDelivery test suite configuration and reset the current test suite and all of its test case progress, select the “RESET TEST SUITE” button (Figure 9, label 6) and re-enroll in a new test suite.

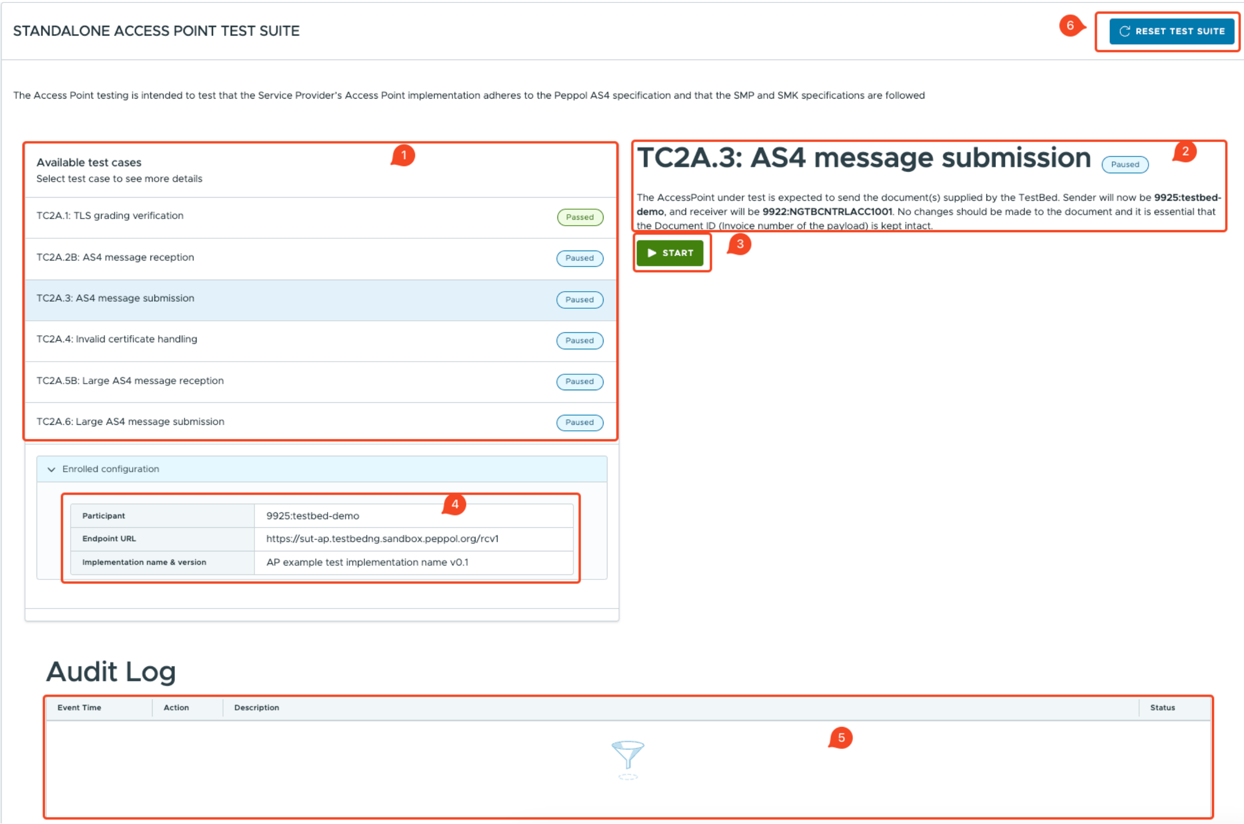

Figure 9 test case overview

The “TLS grading verification” test case performs an automatic grading of the TLS configuration on the provided endpoint URL. The result must be an A or above to fulfill Peppol’s requirements to pass this test. This test case may take up to 15 minutes to complete depending on how complex your infrastructure is to evaluate. The grading is done by a tool provided by Qualys SSL Labs. The grade is A in this example walkthrough supplied endpoint URL; therefore, it passes the test, as seen in Figure 10.

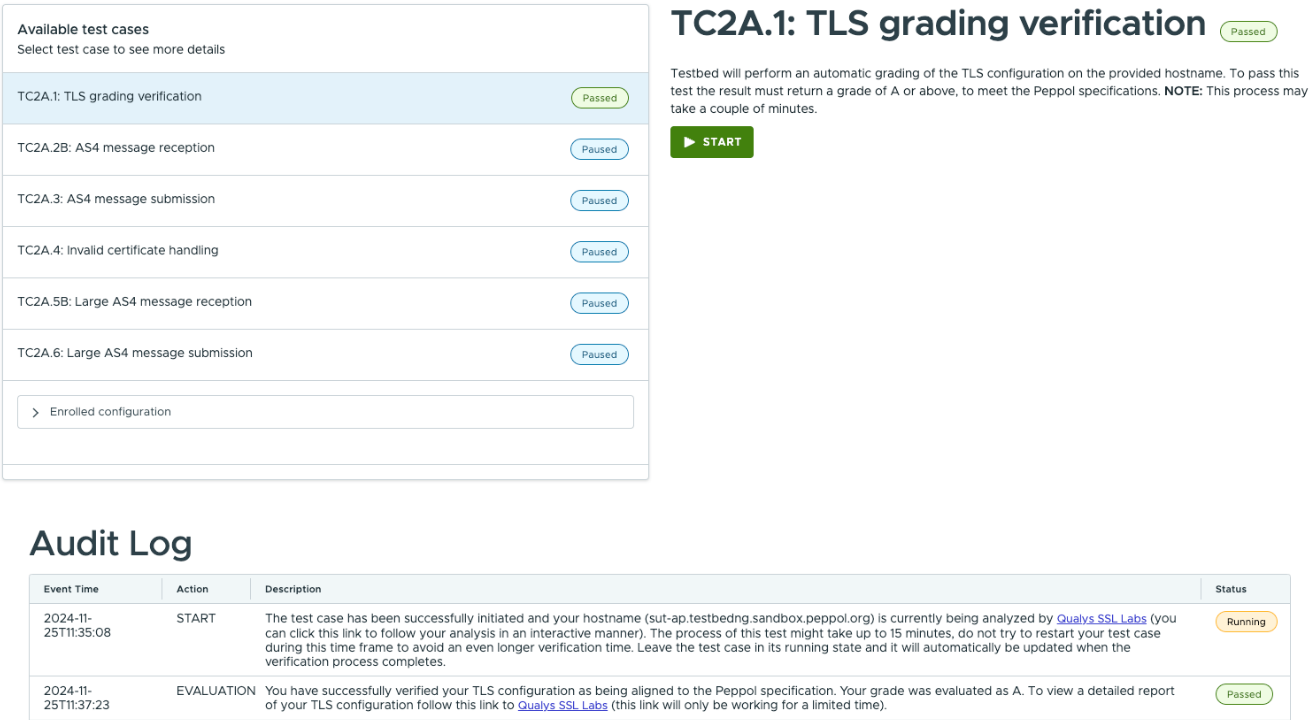

Figure 10 TLS grading example evaluation

In the “AS4 message reception” test case, the Testbed environment will generate a file and send it to the under-test AP. No SMK/SMP participant registration is required, a static configuration will be utilized to address the transaction directly to the AP endpoint URL provided during the AP enrolment form. Figure 11 depicts successfully receiving the message (directly with no SMK/SMP) and passing the test.

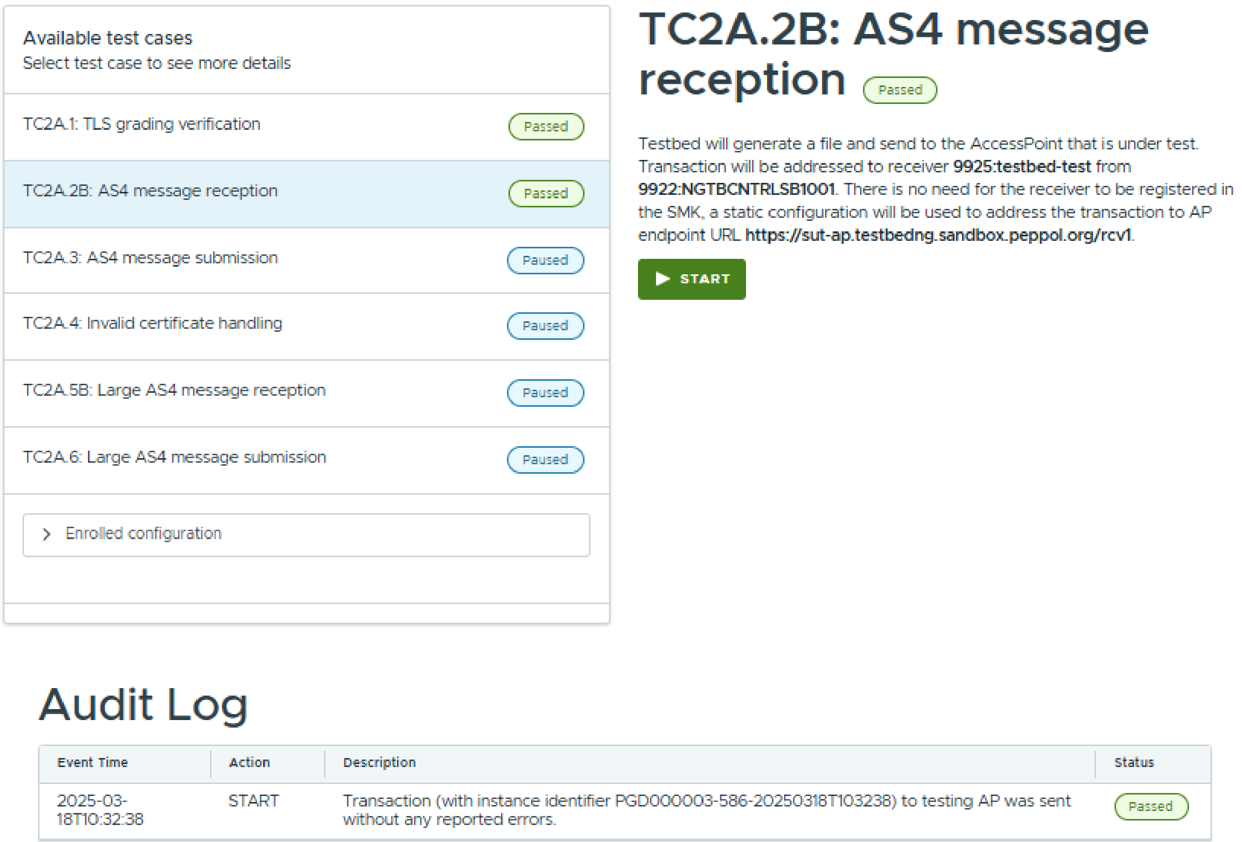

Figure 11 AS4 message reception

In the “AS4 message submission” test case, the under-test AP is expected to send a document supplied by the Testbed environment. The sender will be the participant provided during enrollment (participant id and scheme id), and the receiver will be the Testbed. Download the zip package containing the ready-to-send XML artifact, as shown in

Figure 12. Before sending the file, do not alter it in any way, as the Testbed may have trouble correlating it appropriately.

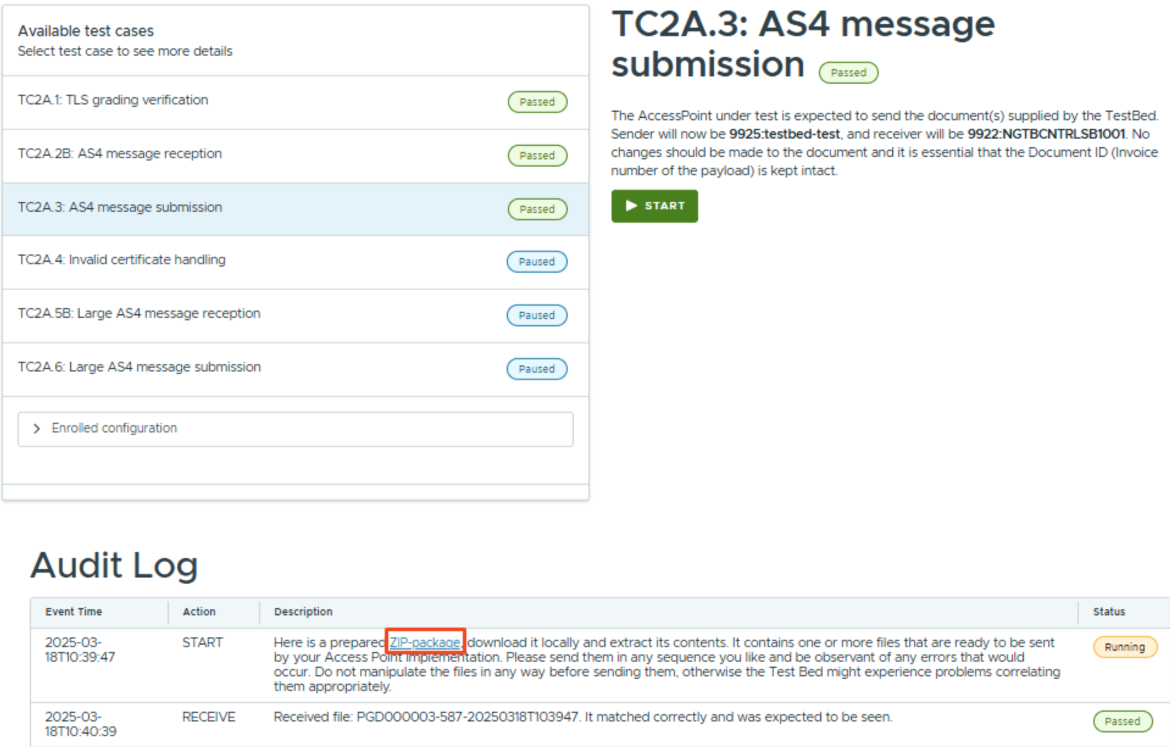

Figure 12 also depicts successfully passing the test. In case of errors, examine them, and once corrected, restart the test.

Figure 12 AS4 Message submission

The “Invalid certificate handling” verifies that your implementation can identify invalid certificates. Download the zip package, containing ready-to-send XML artifacts. The zip package includes 3 XML artifacts to be sent. Please send them in the same sequence as their naming suggests and be observant if any errors occur. Before sending the file, do not alter it in any way, as the Testbed may have trouble correlating it appropriately. Figure 13 shows the successful test scenario. If any errors occur, examine them, and once corrected, restart the test case.

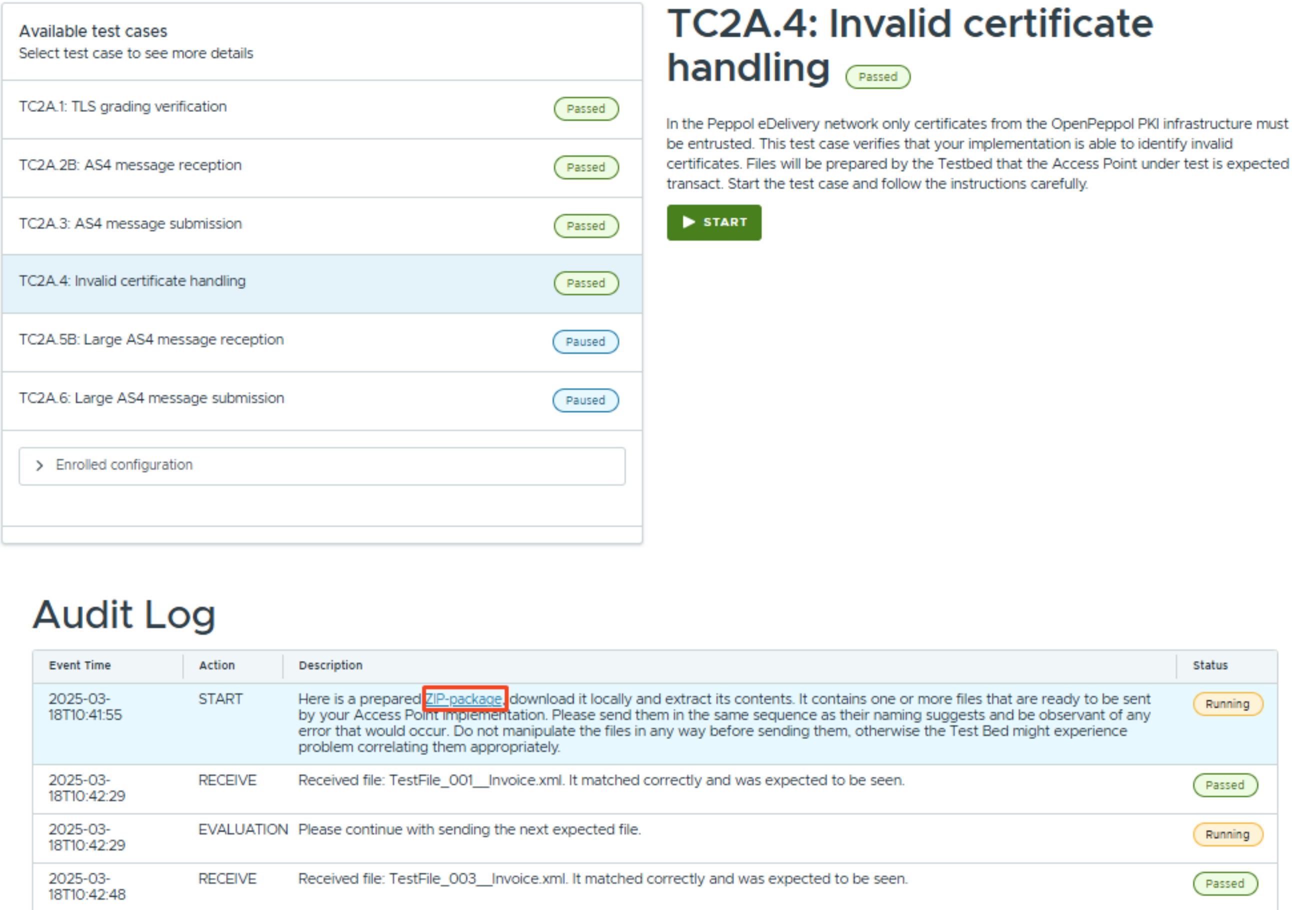

Figure 13 Invalid certificate handling

In the “Large AS4 message reception” test case, the Testbed environment will generate a large file and send it to the under-test AP. No SMK/SMP participant registration is required, a static configuration will be utilized to address the transaction directly to the AP endpoint URL provided during the AP enrolment form. Figure 14 Large AS4 message reception exampledepicts successfully receiving the message (directly with no SMK/SMP) and passing the test.

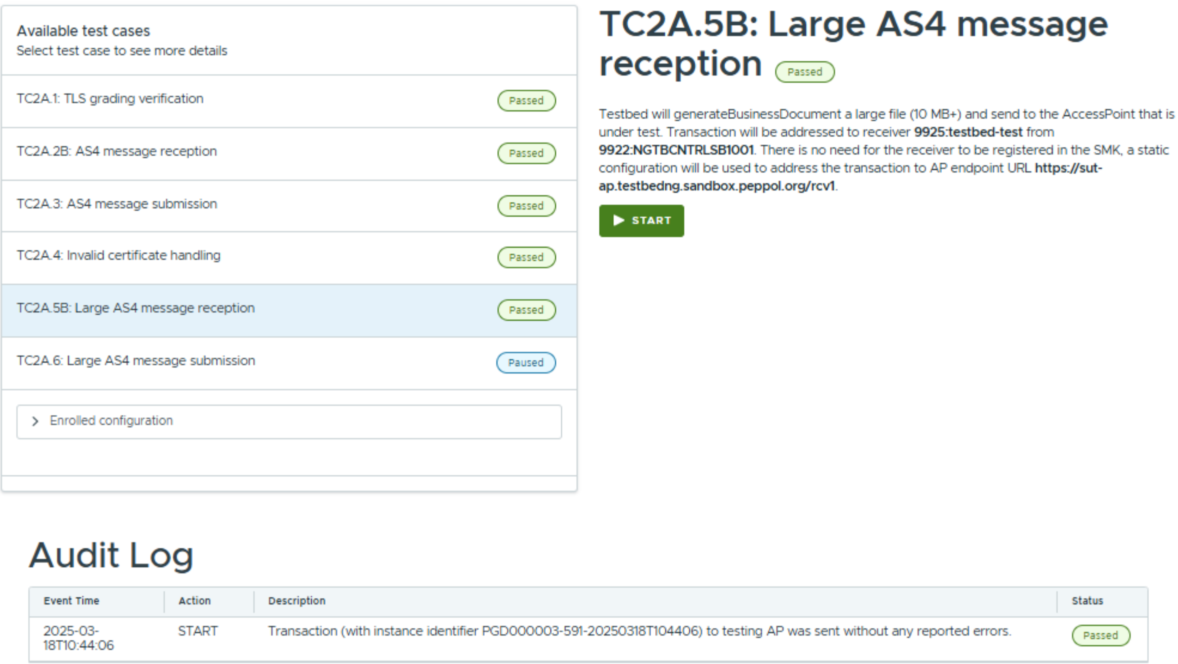

Figure 14 Large AS4 message reception example

In the “Large AS4 message submission” test case, the under-test AP is expected to send a large document supplied by the Testbed environment. The sender will be the participant provided during enrolment (participant id and scheme id), and the receiver will be the Testbed. Download the zip package containing the ready-to-send XML artifact, as shown in Figure 15. Before sending the file, do not alter it in any way, as the Testbed may have trouble correlating it appropriately. Figure 15 also depicts successfully passing the test. In case of errors, examine them, and once corrected, restart the test. When the test suite is complete (all test cases are passed), the user may also download a PDF report, as shown in Figure 15.

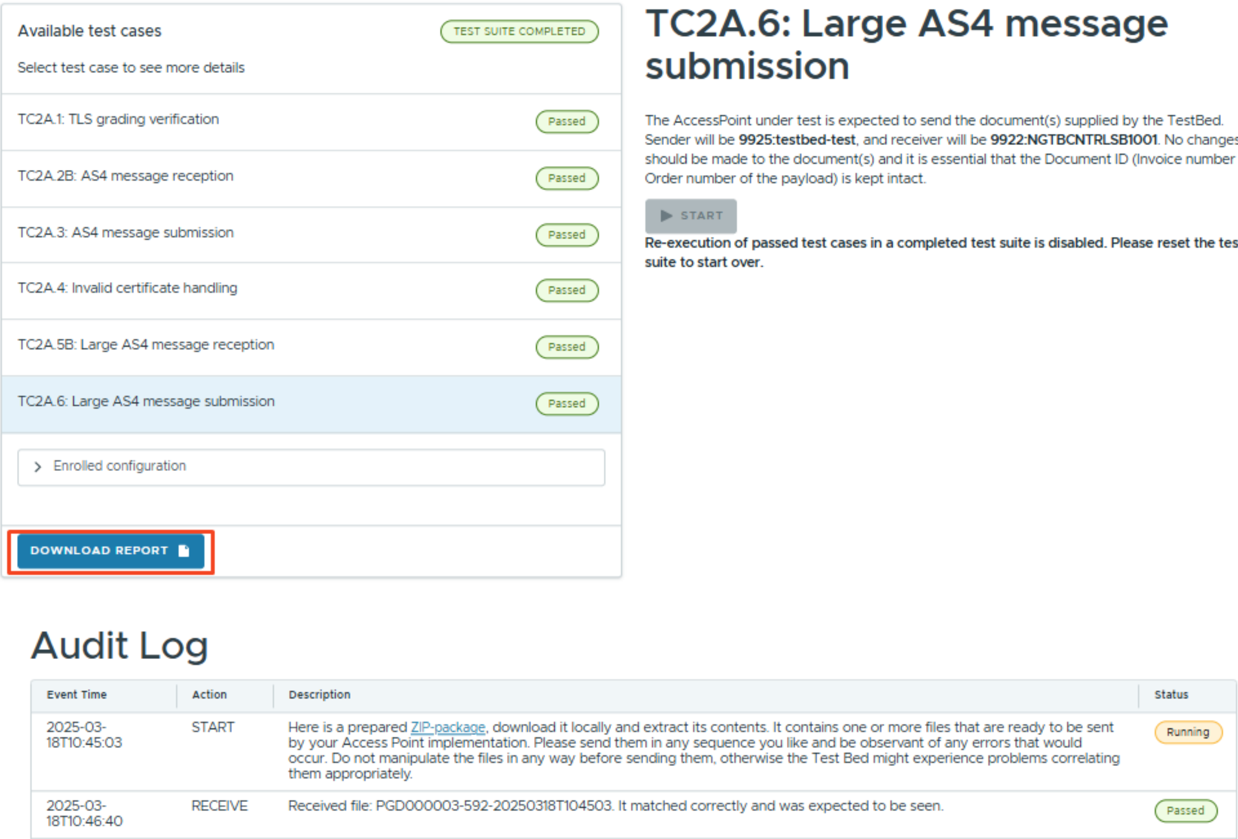

Figure 15 Large AS4 message submission example